Desktop Application Development

TOPIC: Apartment Business Management

Class: B05E

Giảng viên: Hoàng Văn Hiếu

GROUP MEMBER

Nguyễn Văn Việt	225054137
Lê Phi Anh	225054039
Nguyễn Tuấn Hưng	225053771

TABLE OF CONTENTS

- Chapter I: Introduction
  - 1. Abstracts
  - 2. Tool & Function
  - 3. Sketch
- Chapter II: Database
  - 1. Data table
  - 2. Entity Relationship Diagram
- Chapter III: List of Image
- Chapter IV: Upcoming features and Conclusion

CHAPTER 1: INTRODUCTION

1. ABSTRACTS

The service industry is rapidly growing, with many hotels and motels being established to meet the needs of tourism and relaxation.

Problem: Management in motels and hotels is becoming increasingly complex, requiring an effective management solution.

Objective: Develop simple and comprehensive apartment management software with practical application.

CHAPTER 1: INTRODUCTION

2. TOOL & FUNCTIONS

Tools used: Window form, C#, SQL Server, Figma

Reasons for choosing the tool:

+ Widely used in small and medium projects, helping students easily adapt to the professional environment.
+ Both Visual Studio (Community Edition) and MySQL are free.
+ Support for practicing object-oriented programming, web development and database management
+ With Figma, multiple people can work on the same project at the same time, which is useful for group assignments or team projects. And easily create interactive prototypes to simulate how your app or website works.

CHAPTER 1: INTRODUCTION

Functions:

+ Login/Register: Allows users to create an account and log in to the program.

+ Room management

+ Tenant management

+ Contract management

+ Electricity and Water management

+ Manage Invoices: Can issue monthly room invoices

CHAPTER 1: INTRODUCTION

Functions:

+ Statistics: Calculate statistics of electricity bills, water bills, room bills, other services and total monthly revenue.

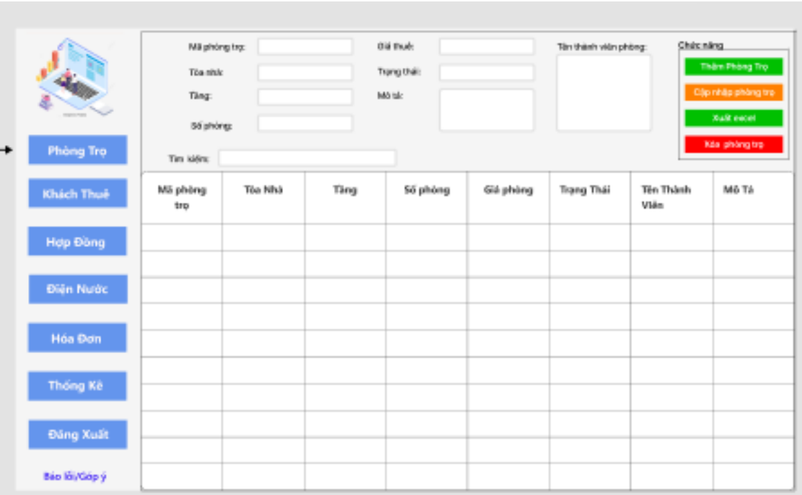
+ Add, edit, delete, export to excel, search function are available on most management pages depending on related information.

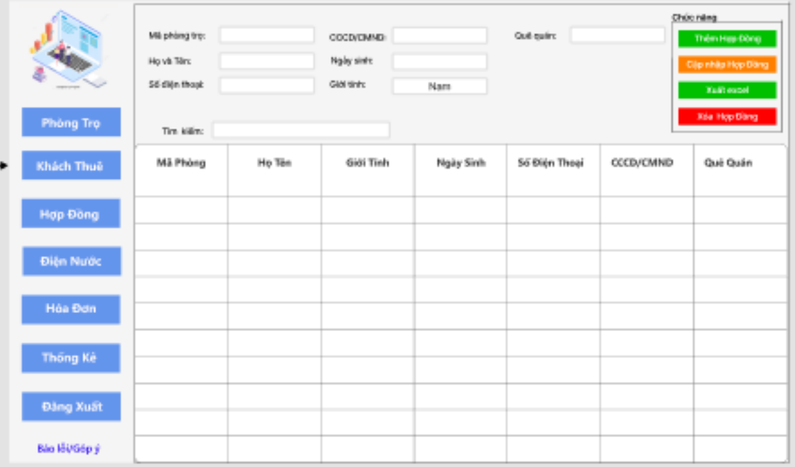
+ Change Password

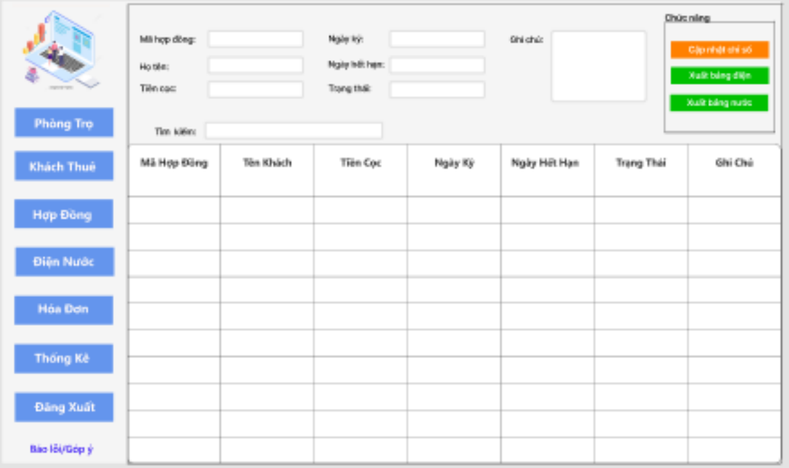
+ Log out: Exit program.

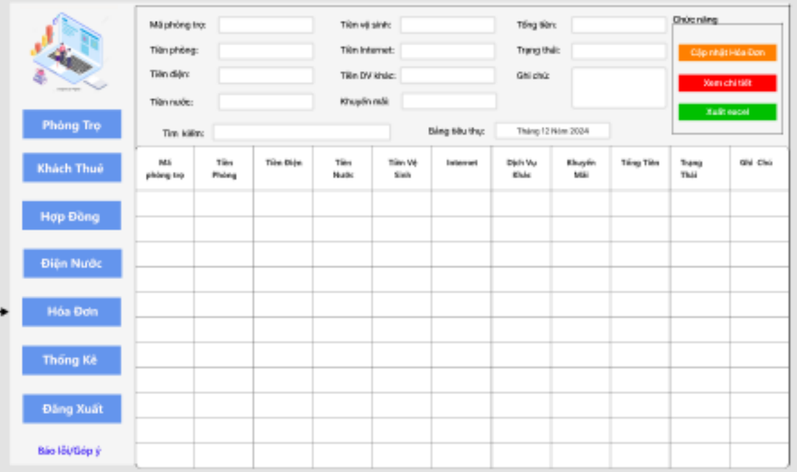
CHAPTER 1: INTRODUCTION

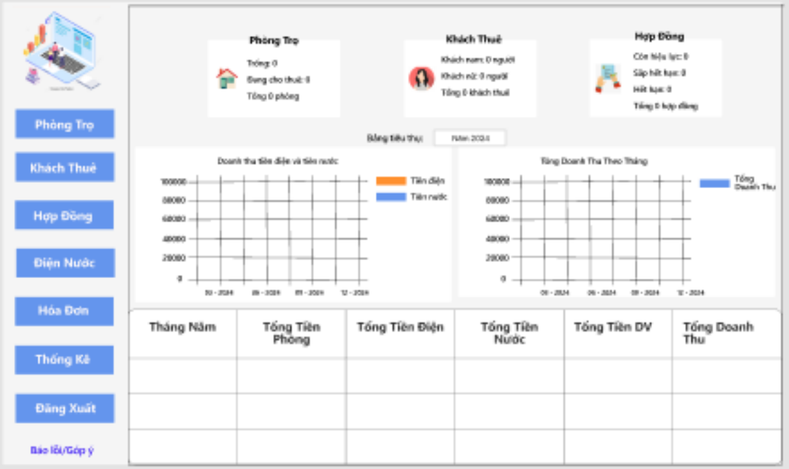
3. Sketch











Chapter II: Database

1. Data Table

a) Login/Register table

Column Name	Data Type	Allow Nulls
Acc	int	No
TenDangNhap	varchar(50)	Yes
MatKhau	varchar(50)	Yes
Email	varchar(100)	Yes

Acc: Primary Key identifies account

Chapter II: Database

1. Data Table

b)Manage Rooms table

Column Name	Data Type	Allow Nulls
MaPhongTro	nvarchar(50)	No
ToaNha	nvarchar(50)	Yes
Tang	int	Yes
SoPhong	int	Yes
GiaPhong	int	Yes
MoTa	nvarchar(MAX)	Yes

MaPhongTro: Primary Key identifies each room

Chapter II: Database

1. Data Table

c) Manage Tenant information table

Column Name	Data Type	Allow Nulls
MaKhachThue	int	No
MaPhongTro	nvarchar(50)	Yes
HoTen	nvarchar(100)	Yes
GioiTinh	nvarchar(10)	Yes
NgaySinh	date	Yes
QueQuan	nvarchar(100)	Yes
SoDienThoai	char(20)	Yes
CCCD	char(50)	Yes

MaKhachThue: The primary key that identifies the tenant.

MaPhongTro: Foreign Key links to the tblPhongTro table, indicating which room the tenant belongs to.

Chapter II: Database

1. Data Table

d) Manage Contracts table

Column Name	Data Type	Allow Nulls
MaPhongTro	nvarchar(50)	Yes
MaHopDong	nvarchar(100)	No
MaKhachThue	int	Yes
HoTenKhachThue	nvarchar(100)	Yes
TienCoc	int	Yes
NgayKy	date	Yes
NgayHetHan	date	Yes
GhiChu	nvarchar(MAX)	Yes

MaHopDong: Primary Key identifying the contract.

MaPhongTro: Foreign Key linking to tblPhongTro.

MaKhachThue: Foreign Key linking to tblKhachThue.

Chapter II: Database

1. Data Table

e) Manage Invoices table

Column Name	Data Type	Allow Nulls
MaHoaDon	int	
MaPhongTro	nvarchar(50)	✓
ThangNam	date	
TienVeSinh	int	
Internet	int	
DichVuKhac	int	
KhuyenMai	int	
GhiChu	nvarchar(MAX)	✓
TrangThai	nvarchar(20)	✓

MaHoaDon: Primary Key identifying the invoice.

MaPhongTro: Foreign Key linking to tblPhongTro.

Chapter II: Database

1. Data Table

f) Electricity and water table

Column Name	Data Type	Allow Nulls
MaDienNuoc	int	
MaPhongTro	nvarchar(50)	✓
ThangNam	date	✓
SoDienCu	int	✓
SoDienMoi	int	✓
TieuThuDien		✓
TienDien		✓
SoNuocCu	int	✓
SoNuocMoi	int	✓
TieuThuNuoc		✓
TienNuoc		✓
DonGiaDien	int	✓
DonGiaNuoc	int	✓

MaDienNuoc: Primary Key identifying the electricity and water record.

MaPhongTro: Foreign Key linking to tblPhongTro.

Chapter II: Database

1. Data Table

g) Manage Invoices Details table

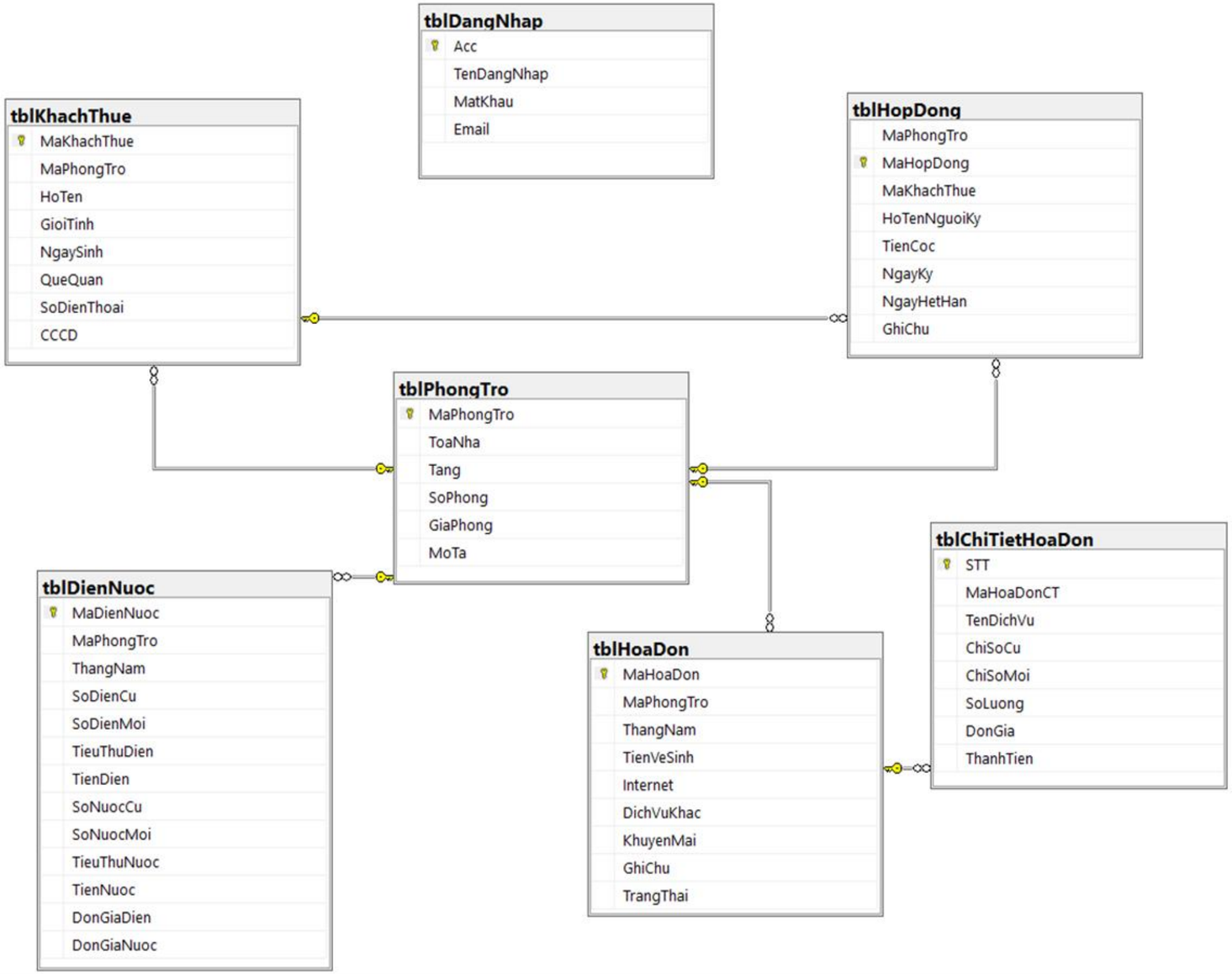
Column Name	Data Type	Allow Nulls
STT	int	
MaHoaDonCT	int	✓
TenDichVu	nvarchar(50)	✓
ChiSoCu	int	✓
ChiSoMoi	int	✓
SoLuong	int	✓
DonGia	int	✓
ThanhTien	int	✓

STT: Primary Key identifies each item.

MaHoaDonCT: Foreign Key links to tblHoaDon.

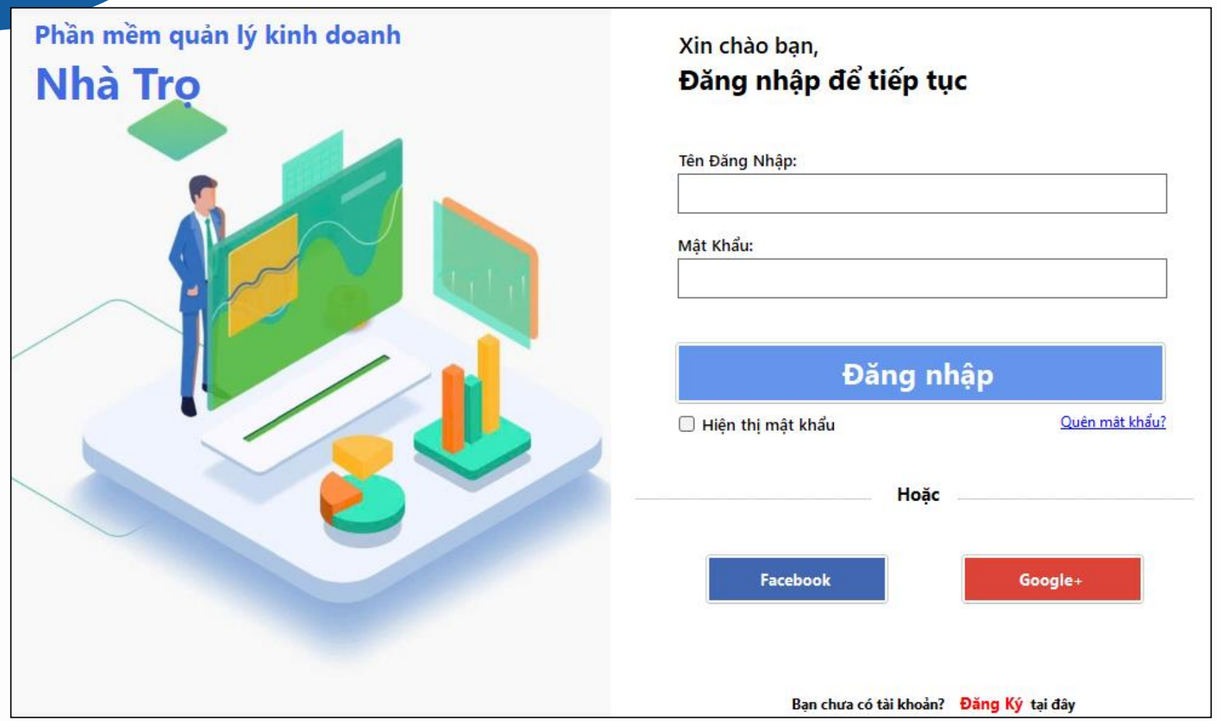
Chapter II: Database

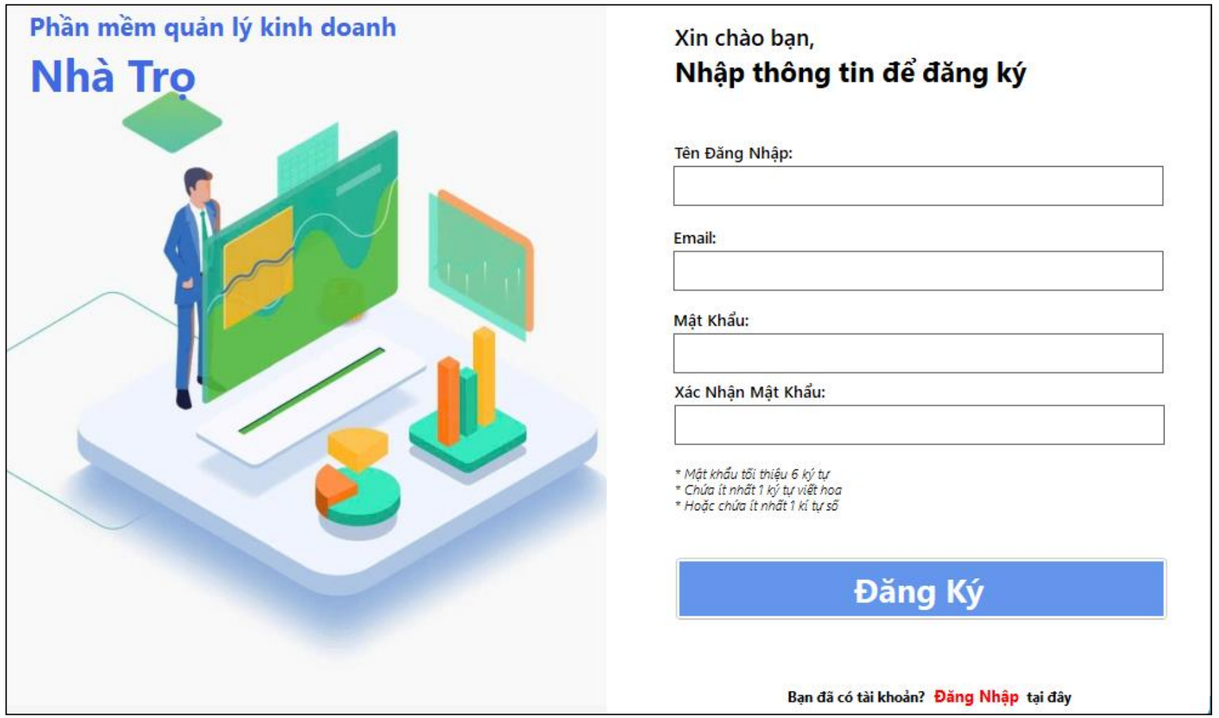
2. Entity Relationship Diagram



CHAPTER III: LIST OF IMAGE

1. Login/Register

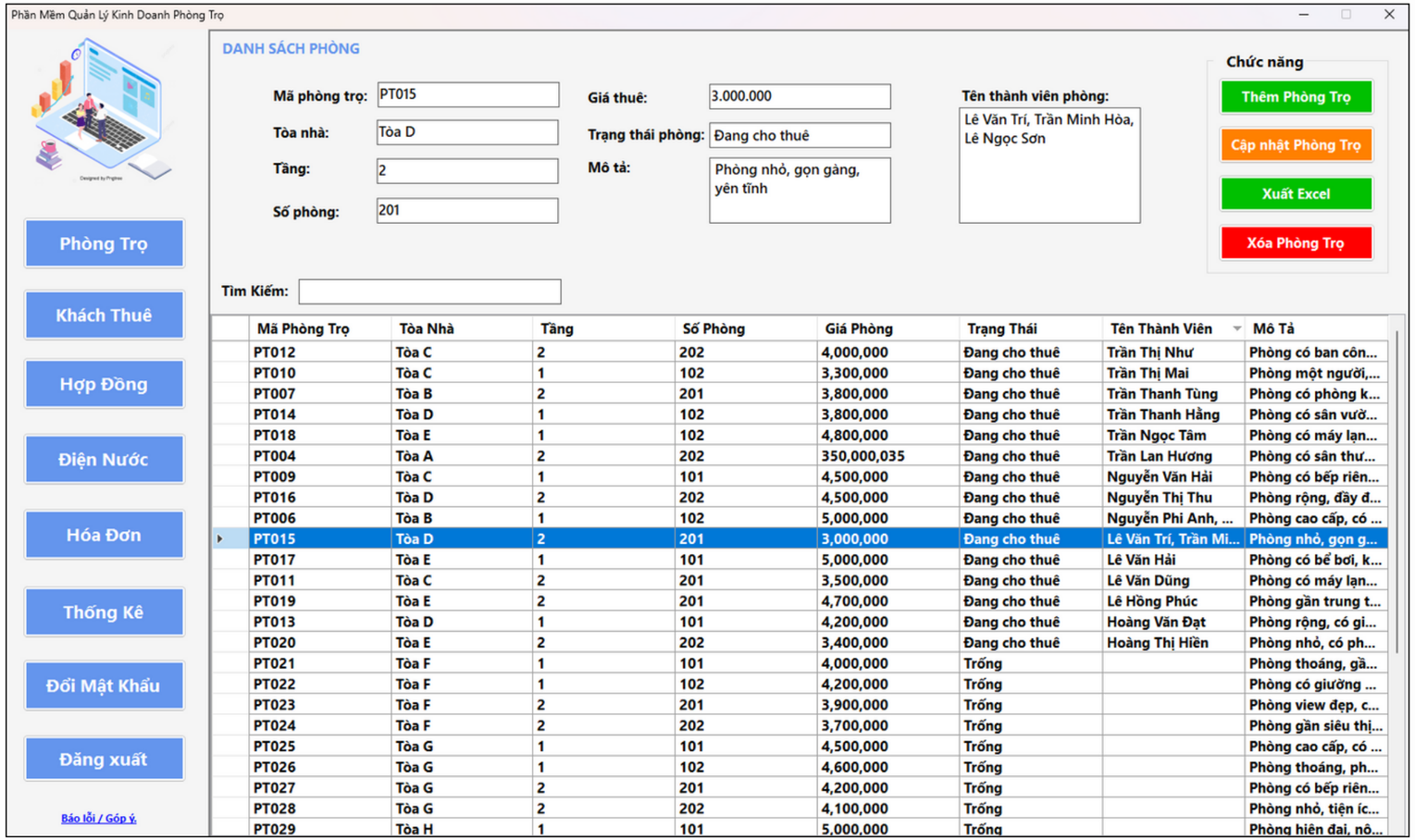




Users must create an account to be able to log in to the program and the password when creating an account must have numbers and uppercase characters.

CHAPTER III: LIST OF IMAGE

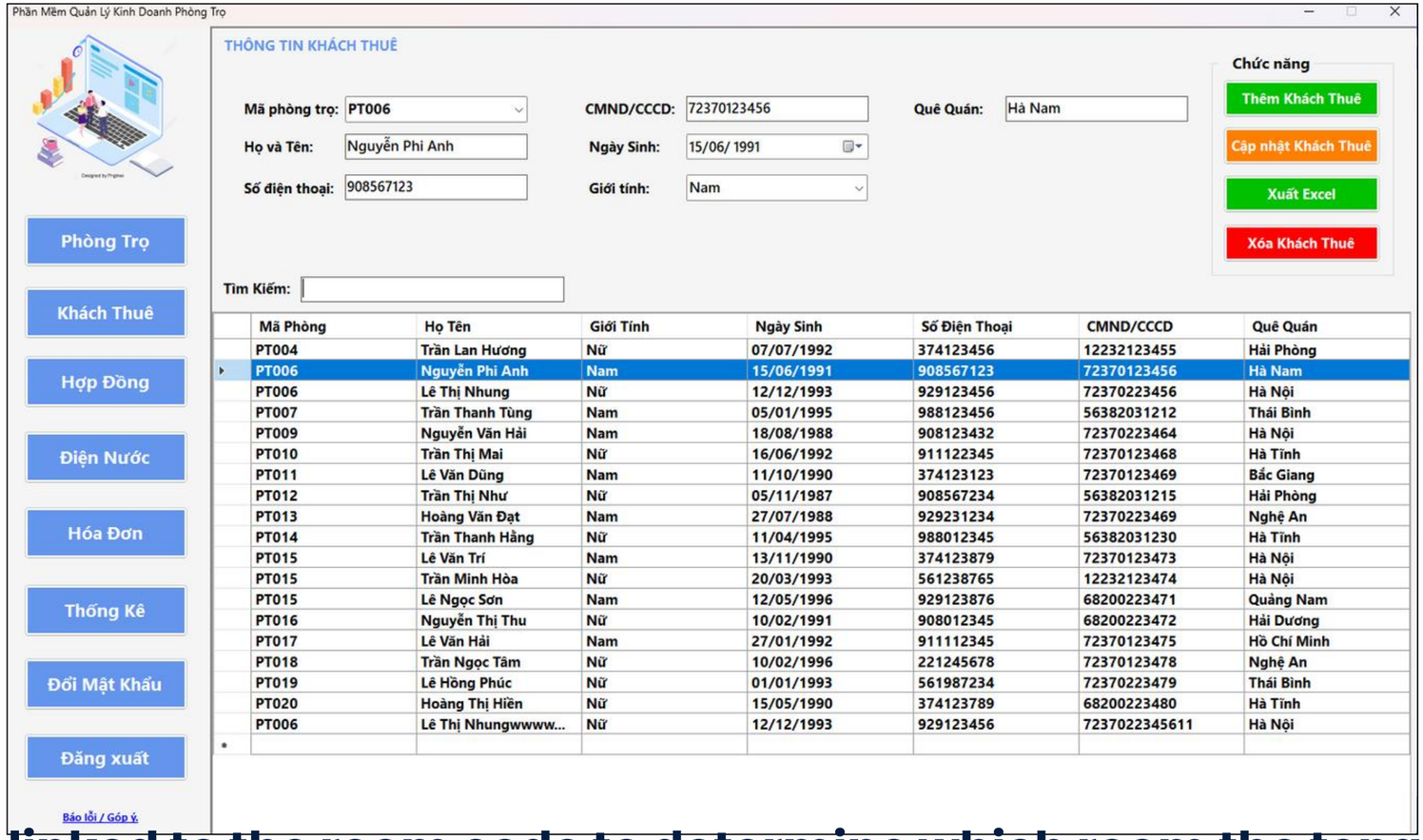
2. Manage Rooms



If the room status is occupied, it is currently rented, otherwise it is vacant. The room member box is easy to manage and see who is in the room to quickly find information.

CHAPTER III: LIST OF IMAGE

3. Manage Tenant information



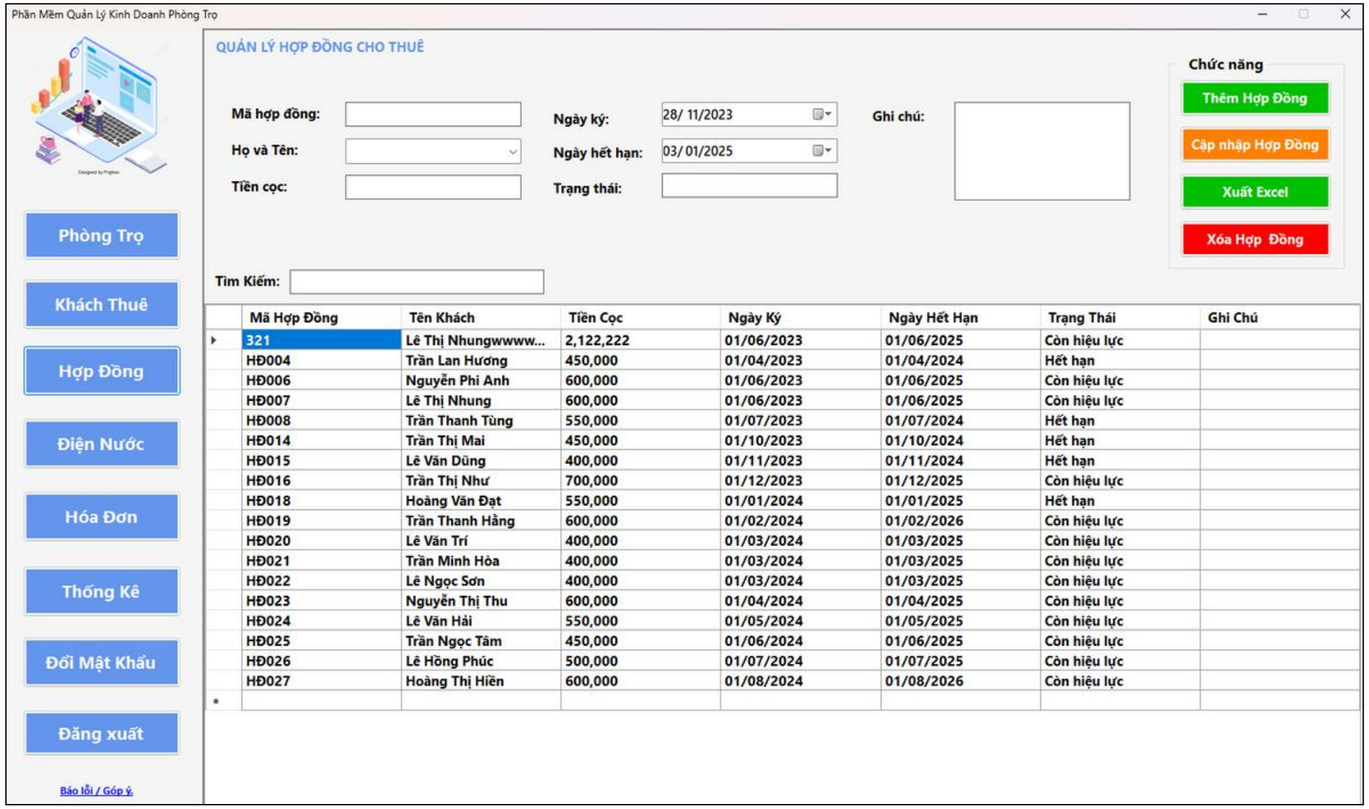
Mã Phòng	Họ Tên	Giới Tính	Ngày Sinh	Số Điện Thoại	CMND/CCCD	Quê Quán
PT004	Trần Lan Hương	Nữ	07/07/1992	374123456	12232123455	Hải Phòng
PT006	Nguyễn Phi Anh	Nam	15/06/1991	908567123	72370123456	Hà Nam
PT006	Lê Thị Nhung	Nữ	12/12/1993	929123456	72370223456	Hà Nội
PT007	Trần Thanh Tùng	Nam	05/01/1995	988123456	56382031212	Thái Bình
PT009	Nguyễn Văn Hải	Nam	18/08/1988	908123432	72370223464	Hà Nội
PT010	Trần Thị Mai	Nữ	16/06/1992	911122345	72370123468	Hà Tĩnh
PT011	Lê Văn Dũng	Nam	11/10/1990	374123123	72370123469	Bắc Giang
PT012	Trần Thị Như	Nữ	05/11/1987	908567234	56382031215	Hải Phòng
PT013	Hoàng Văn Đạt	Nam	27/07/1988	929231234	72370223469	Nghệ An
PT014	Trần Thanh Hằng	Nữ	11/04/1995	988012345	56382031230	Hà Tĩnh
PT015	Lê Văn Trí	Nam	13/11/1990	374123879	72370123473	Hà Nội
PT015	Trần Minh Hòa	Nữ	20/03/1993	561238765	12232123474	Hà Nội
PT015	Lê Ngọc Sơn	Nam	12/05/1996	929123876	68200223471	Quảng Nam
PT016	Nguyễn Thị Thu	Nữ	10/02/1991	908012345	68200223472	Hải Dương
PT017	Lê Văn Hải	Nam	27/01/1992	911112345	72370123475	Hồ Chí Minh
PT018	Trần Ngọc Tâm	Nữ	10/02/1996	221245678	72370123478	Nghệ An
PT019	Lê Hồng Phúc	Nữ	01/01/1993	561987234	72370223479	Thái Bình
PT020	Hoàng Thị Hiền	Nữ	15/05/1990	374123789	68200223480	Hà Tĩnh
PT006	Lê Thị Nhungwwww...	Nữ	12/12/1993	929123456	7237022345611	Hà Nội

The tenant is linked to the room code to determine which room the tenant is staying in

If the room is deleted, the tenant will also be deleted

CHAPTER III: LIST OF IMAGE

4. Manage Contracts



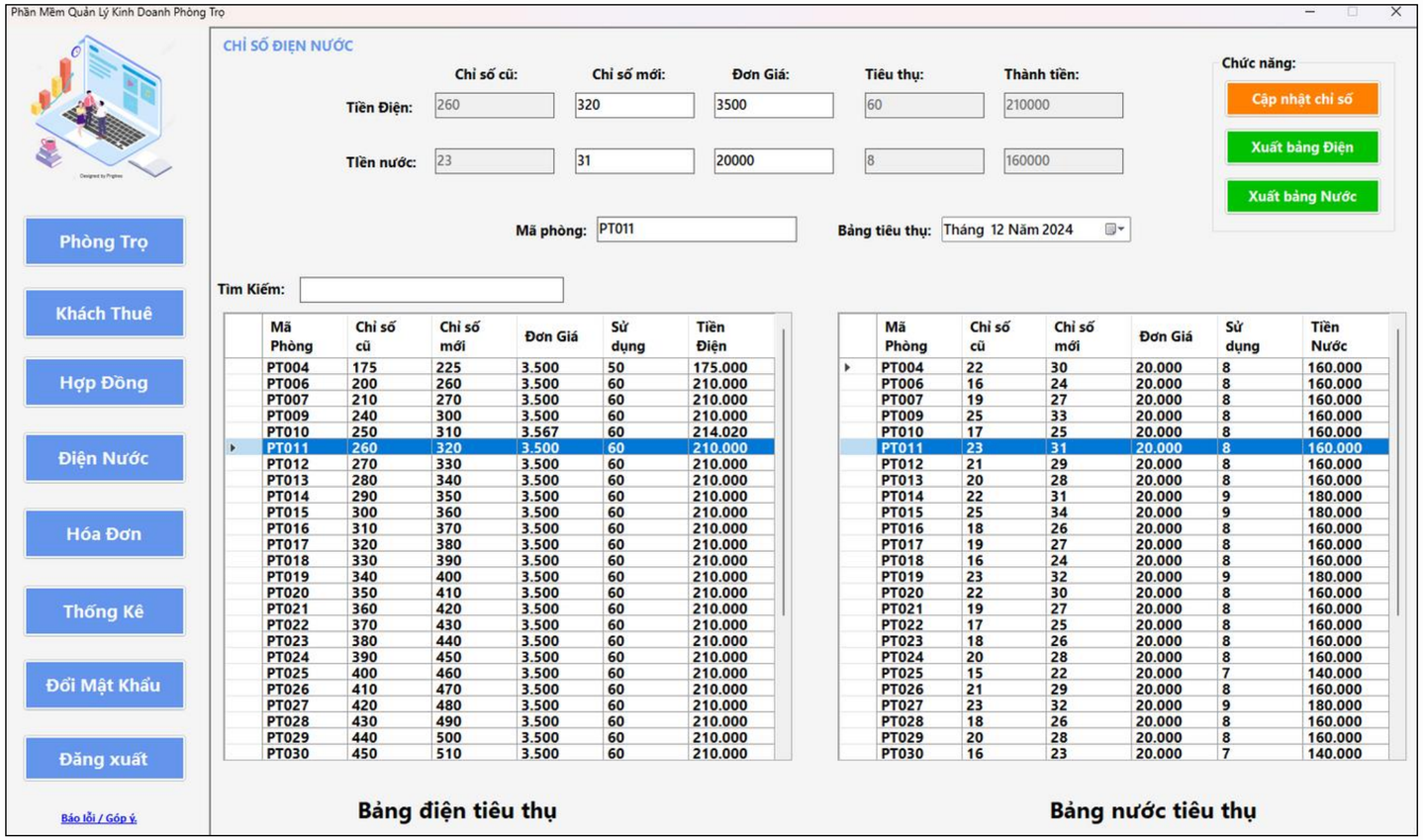
Mã Hợp Đồng	Tên Khách	Tiền Cọc	Ngày Ký	Ngày Hết Hạn	Trạng Thái	Ghi Chú
321	Lê Thị Nhungwwww...	2,122,222	01/06/2023	01/06/2025	Còn hiệu lực	
HĐ004	Trần Lan Hương	450,000	01/04/2023	01/04/2024	Hết hạn	
HĐ006	Nguyễn Phi Anh	600,000	01/06/2023	01/06/2025	Còn hiệu lực	
HĐ007	Lê Thị Nhung	600,000	01/06/2023	01/06/2025	Còn hiệu lực	
HĐ008	Trần Thanh Tùng	550,000	01/07/2023	01/07/2024	Hết hạn	
HĐ014	Trần Thị Mai	450,000	01/10/2023	01/10/2024	Hết hạn	
HĐ015	Lê Văn Dũng	400,000	01/11/2023	01/11/2024	Hết hạn	
HĐ016	Trần Thị Như	700,000	01/12/2023	01/12/2025	Còn hiệu lực	
HĐ018	Hoàng Văn Đạt	550,000	01/01/2024	01/01/2025	Hết hạn	
HĐ019	Trần Thanh Hằng	600,000	01/02/2024	01/02/2026	Còn hiệu lực	
HĐ020	Lê Văn Trí	400,000	01/03/2024	01/03/2025	Còn hiệu lực	
HĐ021	Trần Minh Hòa	400,000	01/03/2024	01/03/2025	Còn hiệu lực	
HĐ022	Lê Ngọc Sơn	400,000	01/03/2024	01/03/2025	Còn hiệu lực	
HĐ023	Nguyễn Thị Thu	600,000	01/04/2024	01/04/2025	Còn hiệu lực	
HĐ024	Lê Văn Hải	550,000	01/05/2024	01/05/2025	Còn hiệu lực	
HĐ025	Trần Ngọc Tâm	450,000	01/06/2024	01/06/2025	Còn hiệu lực	
HĐ026	Lê Hồng Phúc	500,000	01/07/2024	01/07/2025	Còn hiệu lực	
HĐ027	Hoàng Thị Hiền	600,000	01/08/2024	01/08/2026	Còn hiệu lực	

Contract status is based on the expiration date compared to the actual time.

The contract code will follow the tenant, the contract codes cannot be duplicated, the name on the contract must exist in the tenant management to be allowed to add or edit the contract.

CHAPTER III: LIST OF IMAGE

5. Electricity and water management

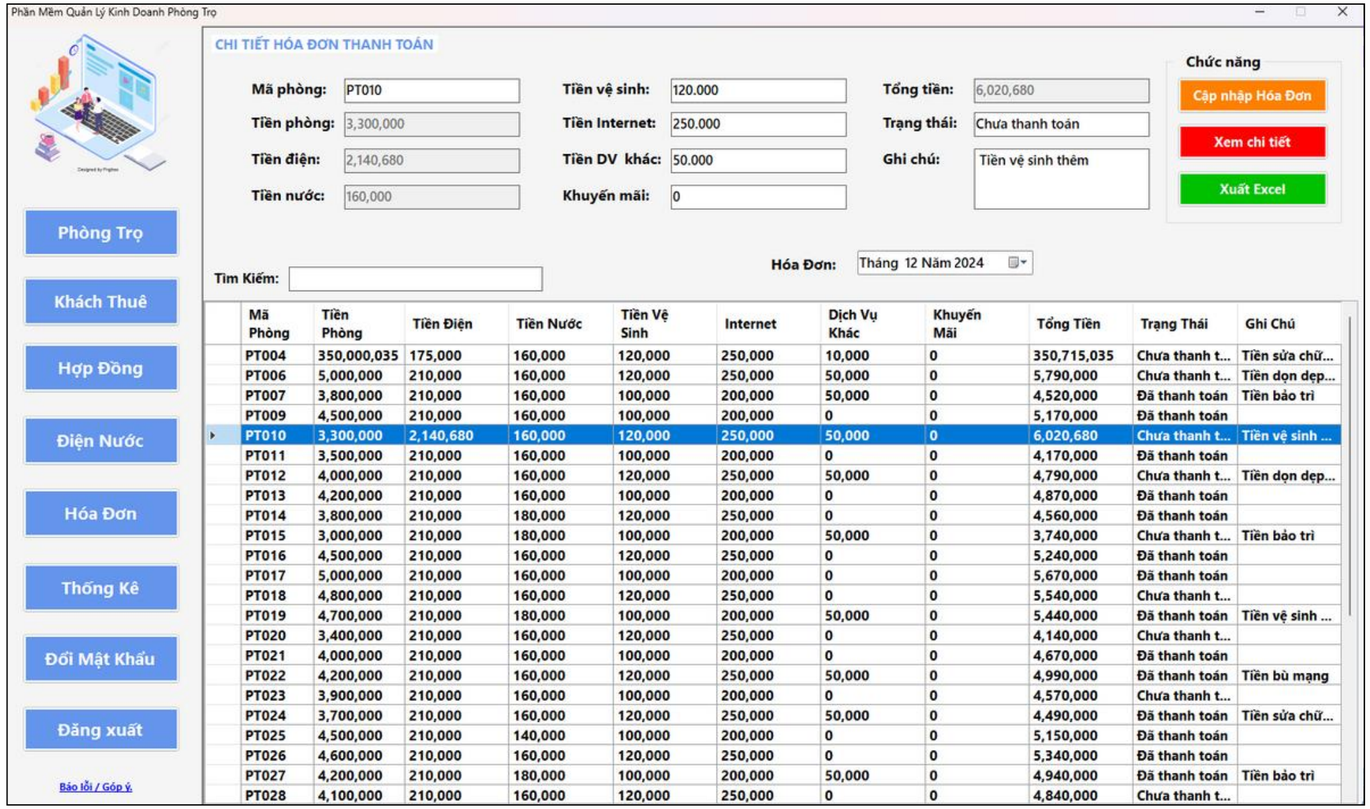


The electricity index from the new index this month to the next month will be inherited to the old index, the unit price for the next month will be inherited from the current month now until the user changes it.

Formula for calculating electricity and water bills = (new index - old index) * unit price

CHAPTER III: LIST OF IMAGE

6. Manage Invoices



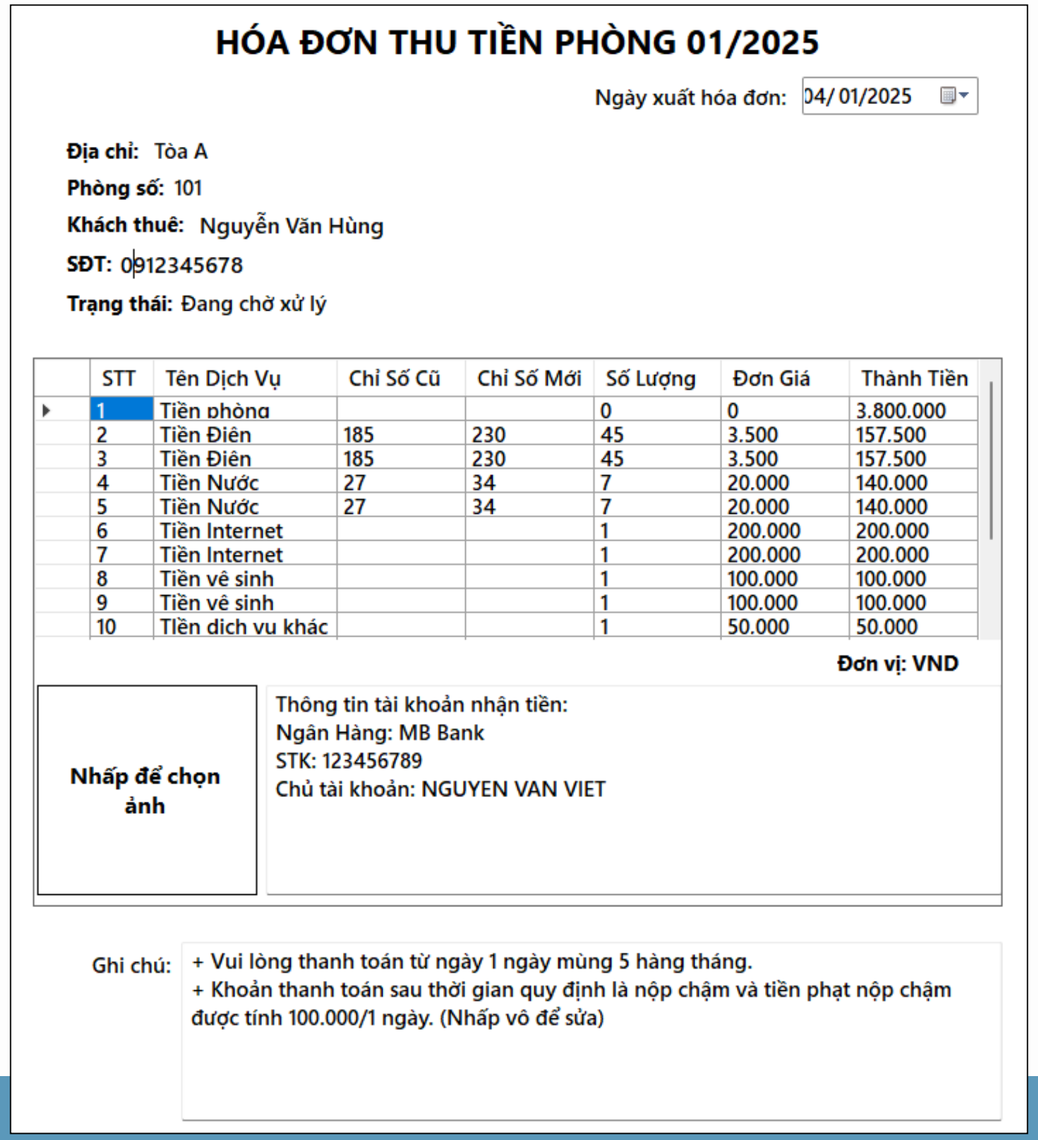
Mã Phòng	Tiền Phòng	Tiền Điện	Tiền Nước	Tiền Vệ Sinh	Internet	Dịch Vụ Khác	Khuyến Mãi	Tổng Tiền	Trạng Thái	Ghi Chú
PT004	350,000,035	175,000	160,000	120,000	250,000	10,000	0	350,715,035	Chưa thanh t...	Tiền sửa chữ...
PT006	5,000,000	210,000	160,000	120,000	250,000	50,000	0	5,790,000	Chưa thanh t...	Tiền dọn dẹp...
PT007	3,800,000	210,000	160,000	100,000	200,000	50,000	0	4,520,000	Đã thanh toán	Tiền bảo trì
PT009	4,500,000	210,000	160,000	100,000	200,000	0	0	5,170,000	Đã thanh toán	
PT010	3,300,000	2,140,680	160,000	120,000	250,000	50,000	0	6,020,680	Chưa thanh t...	Tiền vệ sinh ...
PT011	3,500,000	210,000	160,000	100,000	200,000	0	0	4,170,000	Đã thanh toán	
PT012	4,000,000	210,000	160,000	120,000	250,000	50,000	0	4,790,000	Chưa thanh t...	Tiền dọn dẹp...
PT013	4,200,000	210,000	160,000	100,000	200,000	0	0	4,870,000	Đã thanh toán	
PT014	3,800,000	210,000	180,000	120,000	250,000	0	0	4,560,000	Đã thanh toán	
PT015	3,000,000	210,000	180,000	100,000	200,000	50,000	0	3,740,000	Chưa thanh t...	Tiền bảo trì
PT016	4,500,000	210,000	160,000	120,000	250,000	0	0	5,240,000	Đã thanh toán	
PT017	5,000,000	210,000	160,000	100,000	200,000	0	0	5,670,000	Đã thanh toán	
PT018	4,800,000	210,000	160,000	120,000	250,000	0	0	5,540,000	Chưa thanh t...	
PT019	4,700,000	210,000	180,000	100,000	200,000	50,000	0	5,440,000	Đã thanh toán	Tiền vệ sinh ...
PT020	3,400,000	210,000	160,000	120,000	250,000	0	0	4,140,000	Chưa thanh t...	
PT021	4,000,000	210,000	160,000	100,000	200,000	0	0	4,670,000	Đã thanh toán	
PT022	4,200,000	210,000	160,000	120,000	250,000	50,000	0	4,990,000	Đã thanh toán	Tiền bù mạng
PT023	3,900,000	210,000	160,000	100,000	200,000	0	0	4,570,000	Chưa thanh t...	
PT024	3,700,000	210,000	160,000	120,000	250,000	50,000	0	4,490,000	Đã thanh toán	Tiền sửa chữ...
PT025	4,500,000	210,000	140,000	100,000	200,000	0	0	5,150,000	Đã thanh toán	
PT026	4,600,000	210,000	160,000	120,000	250,000	0	0	5,340,000	Đã thanh toán	
PT027	4,200,000	210,000	180,000	100,000	200,000	50,000	0	4,940,000	Đã thanh toán	Tiền bảo trì
PT028	4,100,000	210,000	160,000	120,000	250,000	0	0	4,840,000	Chưa thanh t...	

Cleaning fees and internet fees are inherited from the current month until user changes

Bill calculation formula = Total of all except promotions

CHAPTER III: LIST OF IMAGE

7. Invoices Details



HÓA ĐƠN THU TIỀN PHÒNG 01/2025

Ngày xuất hóa đơn: 04/ 01/2025

Địa chỉ: Tòa A

Phòng số: 101

Khách thuê: Nguyễn Văn Hùng

SĐT: 0912345678

Trạng thái: Đang chờ xử lý

STT	Tên Dịch Vụ	Chỉ Số Cũ	Chỉ Số Mới	Số Lượng	Đơn Giá	Thành Tiền
1	Tiền phòng			0	0	3.800.000
2	Tiền Điện	185	230	45	3.500	157.500
3	Tiền Điện	185	230	45	3.500	157.500
4	Tiền Nước	27	34	7	20.000	140.000
5	Tiền Nước	27	34	7	20.000	140.000
6	Tiền Internet			1	200.000	200.000
7	Tiền Internet			1	200.000	200.000
8	Tiền vệ sinh			1	100.000	100.000
9	Tiền vệ sinh			1	100.000	100.000
10	Tiền dịch vụ khác			1	50.000	50.000

Đơn vị: VND

Nhấp để chọn ảnh

Thông tin tài khoản nhận tiền:
Ngân Hàng: MB Bank
STK: 123456789
Chủ tài khoản: NGUYEN VAN VIET

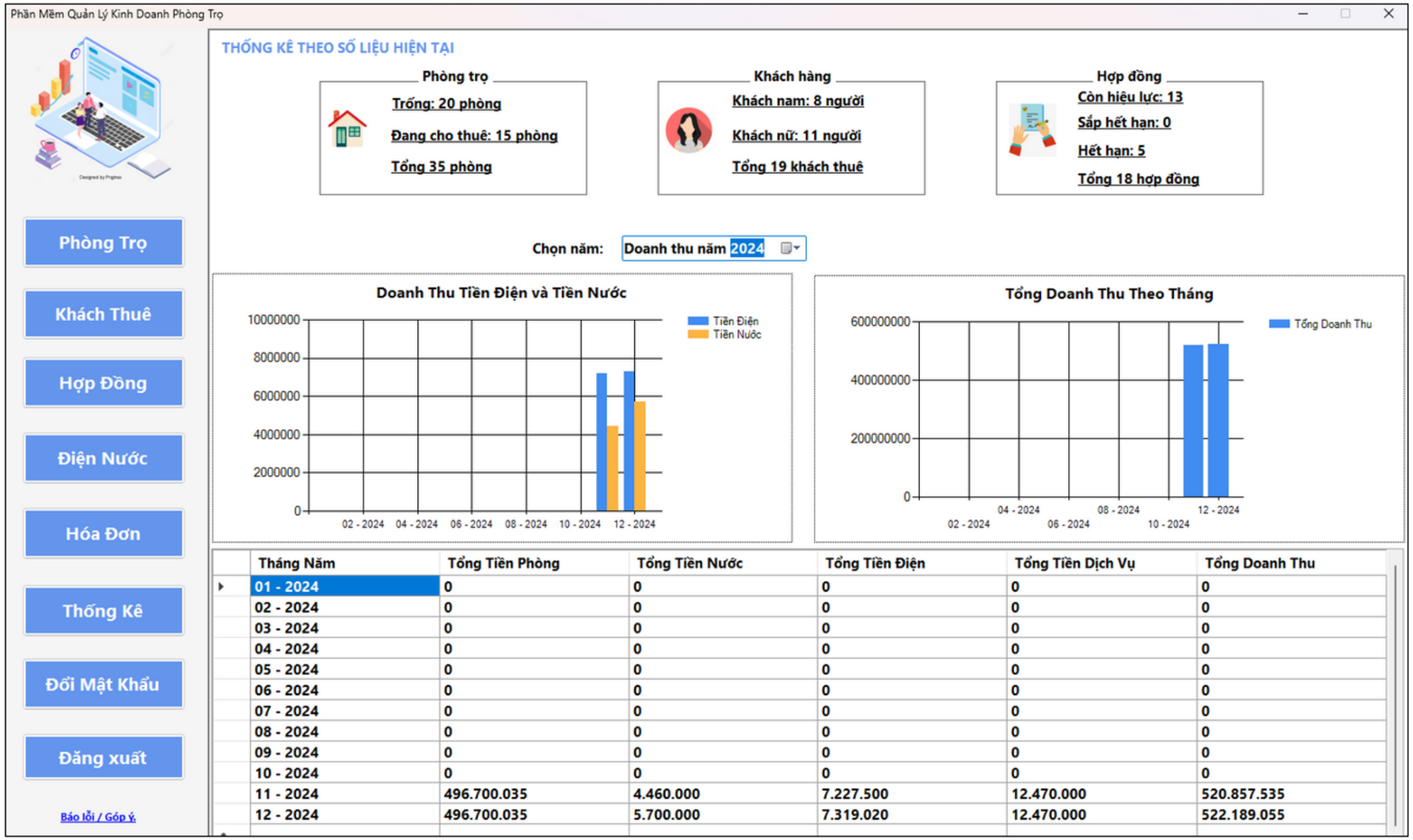
Ghi chú: + Vui lòng thanh toán từ ngày 1 ngày mùng 5 hàng tháng.
+ Khoản thanh toán sau thời gian quy định là nộp chậm và tiền phạt nộp chậm được tính 100.000/1 ngày. (Nhấp vô để sửa)

The details are set up using textboxes so that users can input and change them as appropriate.

The blank box next to the receiving account can be used to select a photo, the user can enter changes, and this photo is saved for the next invoices.

CHAPTER III: LIST OF IMAGE

8. Statistical



Tháng Năm	Tổng Tiền Phòng	Tổng Tiền Nước	Tổng Tiền Điện	Tổng Tiền Dịch Vụ	Tổng Doanh Thu
01 - 2024	0	0	0	0	0
02 - 2024	0	0	0	0	0
03 - 2024	0	0	0	0	0
04 - 2024	0	0	0	0	0
05 - 2024	0	0	0	0	0
06 - 2024	0	0	0	0	0
07 - 2024	0	0	0	0	0
08 - 2024	0	0	0	0	0
09 - 2024	0	0	0	0	0
10 - 2024	0	0	0	0	0
11 - 2024	496.700.035	4.460.000	7.227.500	12.470.000	520.857.535
12 - 2024	496.700.035	5.700.000	7.319.020	12.470.000	522.189.055

- **Statistics of the number of each component on each management page.**
- **Statistics of total income of each month in the year are shown through a data table and two charts of total revenue and electricity and water charts**

CHAPTER III: LIST OF IMAGE

9. Change password

- Validate old password: Requiring the old password to be entered ensures that only the actual user (account owner) can change the password. This reduces the risk of unauthorized password changes.
- Confirm new password: Re-entering the new password to confirm helps avoid password entry errors, ensuring the new password is set correctly.

CHAPTER III: LIST OF IMAGE

10. Export Excel File

Bảng Thanh Toán Hóa Đơn

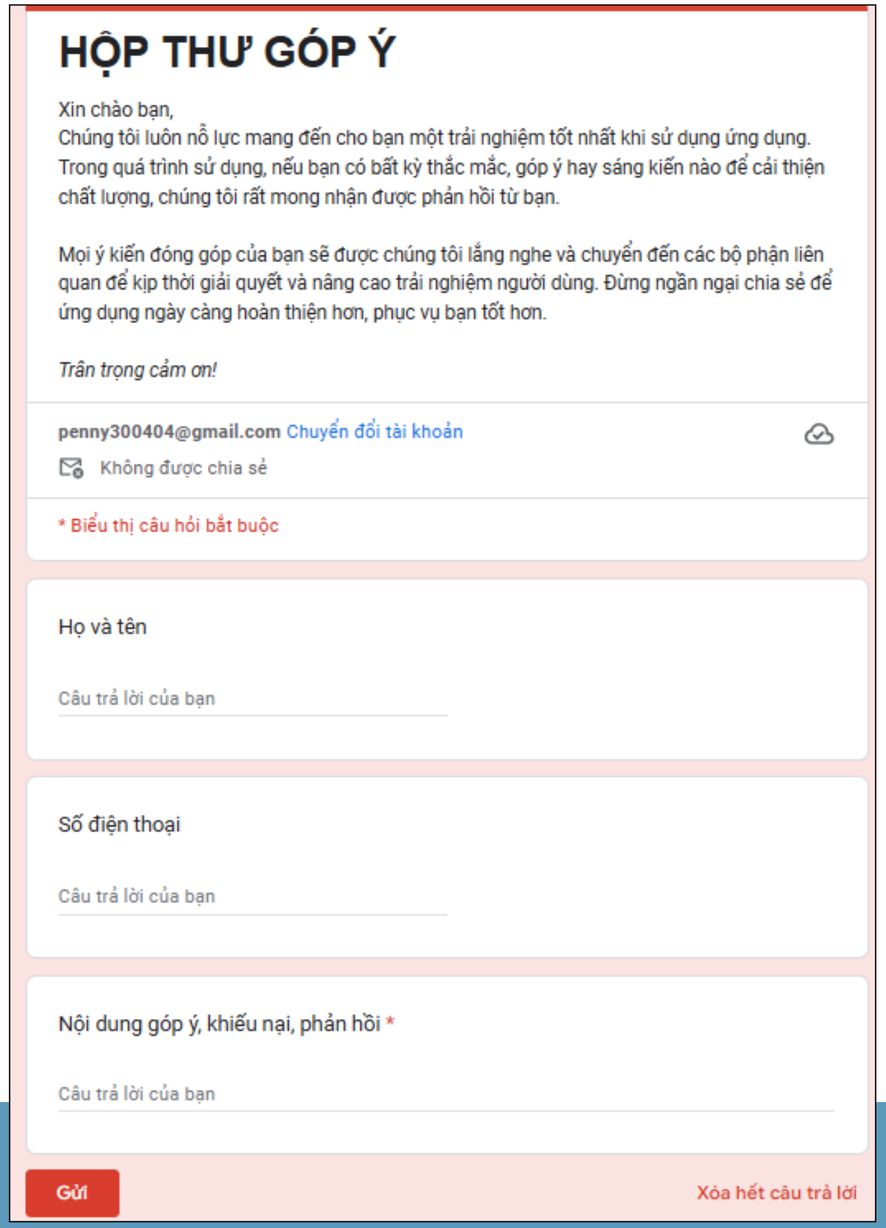
Mã Phòng	Tiền Phòng	Tiền Điện	Tiền Nước	Tiền Vệ Sinh	Internet	Khuyến Mãi	Dịch Vụ Khác	Tổng Tiền	Trạng Thái	Ghi Chú
PT001	3.800.000	157.500	140.000	10.000.033	200.000	50.000	0	14.347.533	Chưa thanh toán	Tiền vệ sinh thêm
PT002	4.200.000	157.500	140.000	120.000	250.000	600.000	0	5.467.500	Đã thanh toán	Tiền bù mạng
PT003	4.600.000	157.500	160.000	100.000	200.000	0	0	5.217.500	Đã thanh toán	
PT004	3.500.000	175.000	160.000	120.000	250.000	10.000	0	4.215.000	Chưa thanh toán	Tiền sửa chữa nhỏ
PT005	3.000.000	175.000	160.000	100.000	200.000	0	0	3.635.000	Đã thanh toán	
PT006	5.000.000	210.000	160.000	120.000	250.000	50.000	0	5.790.000	Chưa thanh toán	Tiền dọn dẹp hàng tháng
PT007	3.800.000	210.000	160.000	100.000	200.000	50.000	0	4.520.000	Đã thanh toán	Tiền bảo trì
PT008	4.000.000	210.000	160.000	120.000	250.000	0	0	4.740.000	Chưa thanh toán	
PT009	4.500.000	210.000	160.000	100.000	200.000	0	0	5.170.000	Đã thanh toán	
PT010	3.300.000	210.000	160.000	120.000	250.000	50.000	0	4.090.000	Chưa thanh toán	Tiền vệ sinh thêm
PT011	3.500.000	210.000	160.000	100.000	200.000	0	0	4.170.000	Đã thanh toán	
PT012	4.000.000	210.000	160.000	120.000	250.000	50.000	0	4.790.000	Chưa thanh toán	Tiền dọn dẹp hàng tháng
PT013	4.200.000	210.000	160.000	100.000	200.000	0	0	4.870.000	Đã thanh toán	
PT014	3.800.000	210.000	180.000	120.000	250.000	0	0	4.560.000	Đã thanh toán	
PT015	3.000.000	210.000	180.000	100.000	200.000	50.000	0	3.740.000	Chưa thanh toán	Tiền bảo trì
PT016	4.500.000	210.000	160.000	120.000	250.000	0	0	5.240.000	Đã thanh toán	
PT017	5.000.000	210.000	160.000	100.000	200.000	0	0	5.670.000	Đã thanh toán	

This function allows users to export excel files of each corresponding management page, so that users can conveniently look up when needed.

CHAPTER III: LIST OF IMAGE

11. Suggestion Box

This suggestion box will collect opinions and complaints about the application from users to help the app know the errors that may occur when users use it, and they will fix the errors to be able to perfect the application.



Chapter IV: UPCOMING FEATURES And Conclusion

1.Login by Google and Facebook

01 Convenient for users: Users do not need to create a new account, just use their existing Google or Facebook account.

02 Higher security: Google and Facebook offer strong security protocols that help protect users' personal information

03 Save development time: Integrating Google and Facebook APIs reduces the time it takes to build an account management system from scratch.

Chapter IV: UPCOMING FEATURES And Conclusion

2. Send invoices to tenants through email

01 Convenient for tenants: Tenants can receive invoices quickly, store them easily and retrieve them at any time.

02 Cost savings: Minimize paper printing and shipping costs.

03 Increase professionalism: Automating the invoice sending process helps improve the reputation and service quality of the boarding house.

Chapter IV: UPCOMING FEATURES And Conclusion

3.Conclusion

01

Through this desktop application course, we gained valuable skills, including how to use essential tools for creating applications, connecting to databases, and formatting tools in Visual Studio.

02

During the project implementation process, we also learned how to solve practical problems, from handling errors in the program to optimizing the performance of the application

THANKS FOR LISTENING!